

Computer Assignment 1 - Incompressible elasticity

FEM-Solids, TME250, 2025

The task is to be carried out **individually** and a report, containing derivations, code (that you have written yourself) and conclusions, is to be handed in via Canvas as a **single pdf file** before **17:00, Friday, November 21, 2025**. The reports, as well as the code, will automatically be sent to Ouriginal, to check for plagiarism. It is okay to discuss the implementation with other students, but **DO NOT** share code with each other. In addition to including the source code in the pdf-file, we ask that you attach your code (e.g. Matlab files) in a zip-file.

A well written report, together with correct implementation, can give maximum 3 points. A late hand in, without prior agreement with the examiner, will give at maximum 2 point. **Contact us in good time if you need to postpone your hand in!**

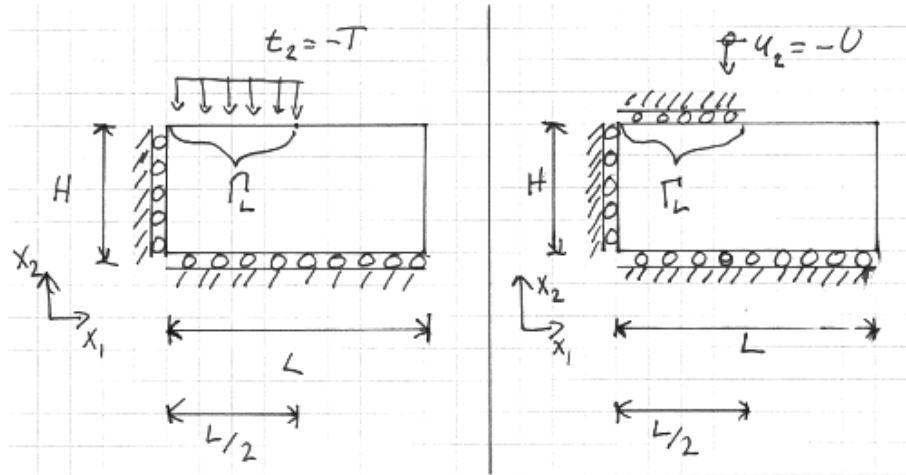


Figure 1: Geometry and load cases: (i) prescribed vertical traction (left) and (ii) prescribed vertical displacement (right). $L = 2H = 40$ mm.

Problem formulation

Consider the two-dimensional domain in the figure, modeling a body under plane strain conditions. It is subjected to symmetric boundary conditions along the left and bottom edge. Part Γ_L of the upper edge is subjected to either (i) prescribed vertical traction or (ii) prescribed vertical displacement, as indicated by Figure 1. The displacements in the horizontal direction are free. Choose suitable values for the Young's modulus E and the amplitude of the prescribed traction T and prescribed displacement U .

The following two scenarios are considered:

- (i) **Uniform traction.** Prescribe a downwards vertical traction $\mathbf{t} = -T\mathbf{e}_2$ all along Γ_L and compute the average displacement

$$\bar{u}_2 = \frac{1}{|\Gamma_L|} \int_{\Gamma_L} \mathbf{e}_2 \cdot \mathbf{u} \, d\Gamma,$$

where $\mathbf{e}_2$ is the unit vector in vertical direction.

Hint: The average displacements can be computed as a matrix multiplication of a (properly) scaled external load vector with the nodal displacements. $\square$

- (ii) **Uniform displacement.** Prescribe a downwards vertical displacement $\mathbf{u} = -U\mathbf{e}_2$ all along Γ_L and compute the reaction in terms of the average vertical traction component

$$\bar{t}_2 = \frac{1}{|\Gamma_L|} \int_{\Gamma_L} \mathbf{e}_2 \cdot \mathbf{t} \, d\Gamma.$$

Here, $\mathbf{e}_2$ is the unit vector in vertical direction.

Hint: The average traction can be computed by adding the pertinent reaction forces and dividing by the length of Γ_L . $\square$

The task is to study the response using different finite element approximations and for different values of Poisson's ratio. In particular, you should study the limit of $\nu \rightarrow 0.5$ pertaining to incompressibility. Implement the solution in a programming language of your choice.

The toolbox CALFEM can be used. If you choose to work in Matlan, the supplied program `mesh.m` generates the topology of a uniform mesh that can be transferred to CALFEM format. (You must do this yourself!)

1 Specific tasks

1. Estimate analytical approximations for the response expressed as functions of ν . Use the approximation of uniform strain (and stress) in the rectangular domain beneath Γ_L .

The traction then becomes $T = -\bar{t}_2 = -\sigma_{22}$ and the vertical displacement can be expressed as $U = -\bar{u}_2 = -\epsilon_{22}H$, where H is the height of the domain.

Consider the two extremes of assuming $\epsilon_{11} = 0$ or $\sigma_{11} = 0$ together with the plane strain assumption to compute approximate relations between T and U as functions of ν , see Figure 2.

2. Use conventional linear triangular elements (CST) and compute the response for varying values of ν , approaching the limit of $\nu \rightarrow 0.5$. For instance you could study $\nu = 0.3, 0.45, 0.49, 0.499, \dots$. Compare the resulting average displacements and average tractions with the respective analytical approximations in a plot.

Now study the convergence of the quantities $\bar{t}_2$ and $\bar{u}_2$ (scenario (i) and (ii), respectively) with mesh-refinement for each selected value of ν . Here, you may use the linear triangular elements existing in the CALFEM toolbox. In particular, plot the pressure distribution (which will be constant in a CST element) for the varying choices of ν .

3. Construct a Taylor-Hood element by combining quadratic approximation of the displacements with linear approximation of the pressure. Repeat the study in the previous task, and compare the results. In addition, consider the case of complete incompressibility ($\nu = 0.5$). Comment on the results.

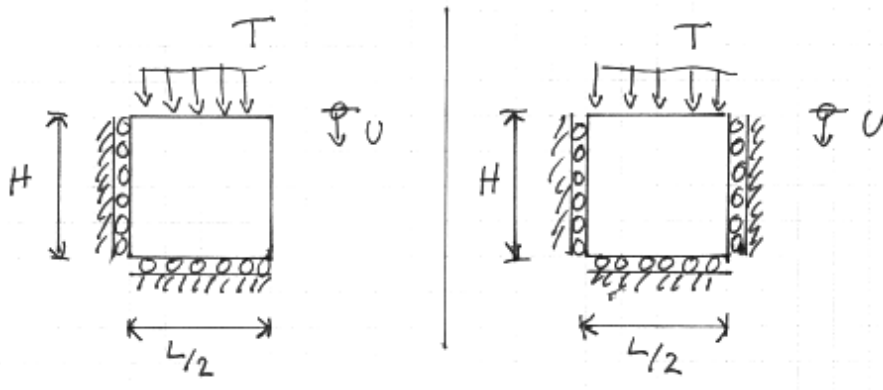


Figure 2: Sketch of the two extremes for analytical estimations: $\sigma_{11} = 0$ (left) and ϵ_{11} (right).

2 Hints

1. Since the top edge, where the load is applied, is parallel to the x -axis the external load vector can simply be integrated by hand. The corresponding shape functions for 1D linear and quadratic elements can be found in e.g. Ottosen and Peterson (1992).
2. To calculate the (average) displacement response on the loaded part of the top boundary, use the established finite element load vector. (Don't just compute it as the mean of the vertical displacement degrees of freedom on boundary segment.)

More specifically, use the fact that

$$\bar{u}_2 = \frac{1}{|\Gamma_L|} \int_{\Gamma_L} \mathbf{e}_2 \cdot \mathbf{u} \, d\Gamma = \sum_k \int_{\Gamma_L} \mathbf{N}_k^{(u)} \cdot \frac{1}{|\Gamma_L|} \mathbf{e}_2 \, d\Gamma (\underline{a}^{(u)})_k = \underline{q}^T \underline{a}^{(u)},$$

where $\underline{q}$ can be obtained by scaling the known part of finite element load vector

$$(\underline{f})_k = \int_{\Gamma_L} \mathbf{N}_k^{(u)} \cdot (-T) \mathbf{e}_2 \, d\Gamma$$

3. To aid in debugging, please check that your element routine generates the following results. The following inputs were used to produce the matrices: $K = 1.25$, $G = 0.4$, thickness $t = 2$, $\mathbf{x}^1 = (6, -4)$, $\mathbf{x}^2 = (5.5, -0.5)$, $\mathbf{x}^3 = (5, 3)$, $\mathbf{x}^4 = (0.5, 1)$, $\mathbf{x}^5 = (-4, -1)$, $\mathbf{x}^6 = (1, -2.5)$. The degrees of freedom are numbered according to Figure 3.2 in the course literature.

CST element:

0.9095	-0.7433	-0.2179	0.4259	-0.6915	0.3174
-0.7433	2.2515	-0.1575	-2.3239	0.9007	0.0724
-0.2179	-0.1575	0.8366	0.6194	-0.6187	-0.4619
0.4259	-2.3239	0.6194	2.7154	-1.0453	-0.3915
-0.6915	0.9007	-0.6187	-1.0453	1.3102	0.1445
0.3174	0.0724	-0.4619	-0.3915	0.1445	0.3192

Taylor-Hood element:

$$K = \begin{bmatrix} K_{uu} & K_{up} \\ K_{up}^T & K_{pp} \end{bmatrix} =$$

0.6109	-0.0716	-0.5891	-0.4272	0.1473	0.1068	0	0	0.0564	-0.1307	-0.2255	0.5227	-1.3333	0	0
-0.0716	0.7403	0.4617	-0.8597	-0.1154	0.2149	0	0	0.0915	0.0318	-0.3662	-0.1274	3.0000	0	0
-0.5891	0.4617	2.2342	0.0027	-0.5891	-0.4272	-0.4511	0.1566	0	0	-0.6050	-0.1937	-3.3333	-3.6667	-2.3333
-0.4272	-0.8597	0.0027	2.5207	0.4617	-0.8597	0.1566	-0.2547	0	0	-0.1937	-0.5466	-3.6667	2.6667	-0.3333
0.1473	-0.1154	-0.5891	0.4617	0.6687	0.0597	-0.3025	-0.5413	0.0756	0.1353	0	0	0	-1.0000	0
0.1068	0.2149	-0.4272	-0.8597	0.0597	0.8498	0.3476	-0.2733	-0.0869	0.0683	0	0	0	-3.3333	0
0	0	-0.4511	0.1566	-0.3025	0.3476	2.2342	0.0027	-0.3025	-0.5413	-1.1781	0.0345	1.3333	3.6667	0.3333
0	0	0.1566	-0.2547	-0.5413	-0.2733	0.0027	2.5207	0.3476	-0.2733	0.0345	-1.7194	-3.0000	-2.6667	-6.3333
0.0564	0.0915	0	0	0.0756	-0.0869	-0.3025	0.3476	0.3960	0.0139	-0.2255	-0.3662	0	0	2.3333
-0.1307	0.0318	0	0	0.1353	0.0683	-0.5413	-0.2733	0.0139	0.3005	0.5227	-0.1274	0	0	0.3333
-0.2255	-0.3662	-0.6050	-0.1937	0	0	-1.1781	0.0345	-0.2255	0.5227	2.2342	0.0027	3.3333	1.0000	-0.3333
0.5227	-0.1274	-0.1937	-0.5466	0	0	0.0345	-1.7194	-0.3662	-0.1274	0.0027	2.5207	3.6667	3.3333	6.3333

-1.3333	3.0000	-3.3333	-3.6667	0	0	1.3333	-3.0000	0	0	3.3333	3.6667	-8.9333	-4.4667	-4.4667
0	0	-3.6667	2.6667	-1.0000	-3.3333	3.6667	-2.6667	0	0	1.0000	3.3333	-4.4667	-8.9333	-4.4667
0	0	-2.3333	-0.3333	0	0	0.3333	-6.3333	2.3333	0.3333	-0.3333	6.3333	-4.4667	-4.4667	-8.9333